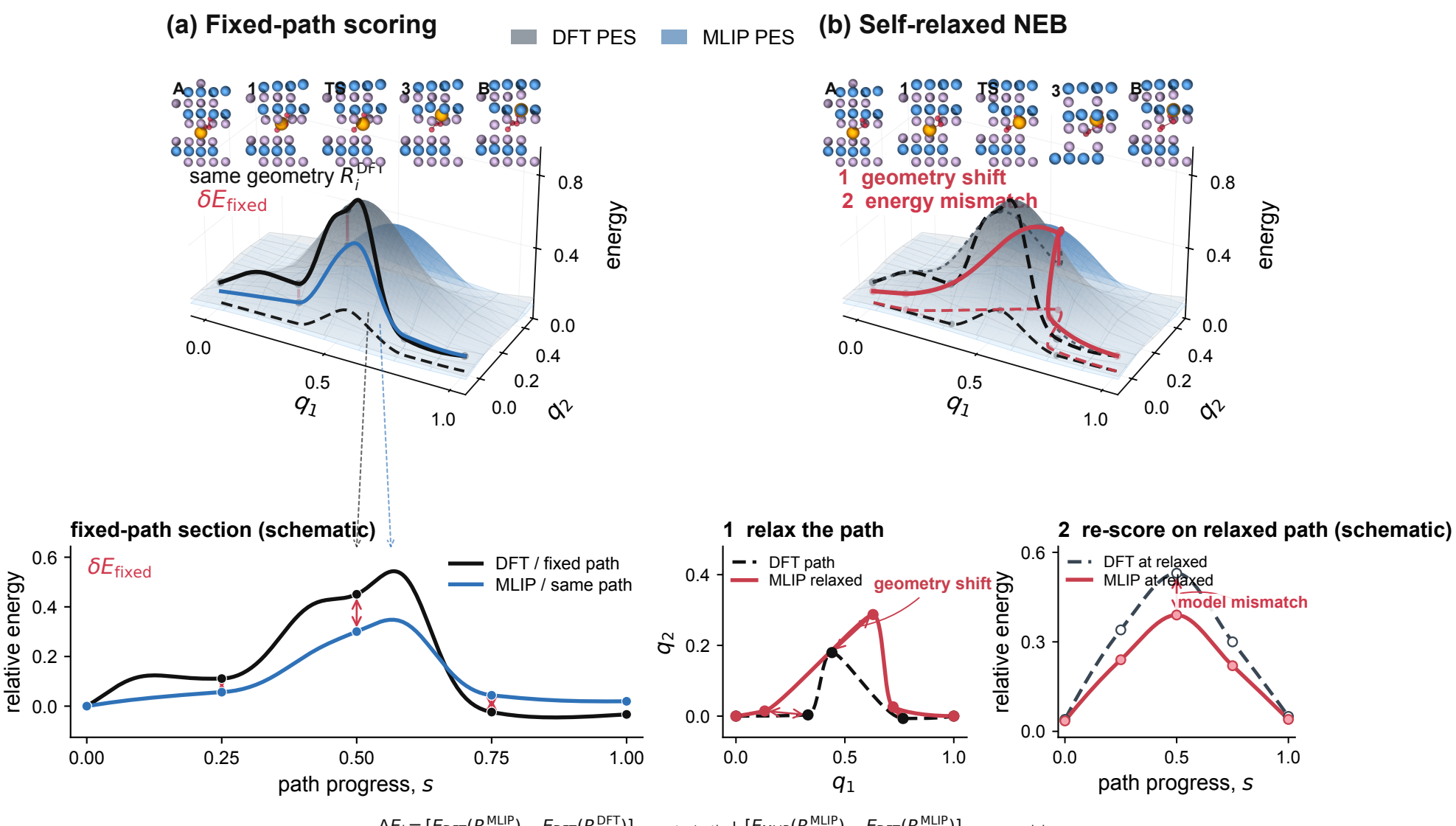

(a) Fixed-path scoring
DFT PES
MLIP PES
(b) Self-relaxed NEB
A
1
TS
3
B
same geometry R_i^{DFT}
δE_{fixed}
1 geometry shift
2 energy mismatch
energy
q_1
q_2
fixed-path section (schematic)
DFT / fixed path
MLIP / same path
relative energy
path progress, s
1 relax the path
DFT path
MLIP relaxed
geometry shift
2 re-score on relaxed path (schematic)
DFT at relaxed
MLIP at relaxed
model mismatch
$\Delta E_i = [E_{DFT}(R_i^{MLIP}) - E_{DFT}(R_i^{DFT})]_{geometry/path} + [E_{MLIP}(R_i^{MLIP}) - E_{DFT}(R_i^{MLIP})]_{energy\ model}$